

AUTOMATIC TEST EQUIPMENT

Becoming Modular



Vertica Asthana
is a technical
journalist at EFY

Widely adopted in highly demanding engineering sectors, automated test equipment (ATE) are now silently making inroads in less capital-intensive sectors like lighting, too. Lighting majors are starting to realise the cost and labour benefits of using ATE in PCB testing.

Manufacturers in defence, avionics, automotive, telecom and robotics industries, on the other hand, use modular ATEs to adhere to the protocols of testing. As the technology evolves, they are asking for more and more features and application updates. The biggest challenge before test and measurement equipment manufacturers, though, remains the growing demand for tools that can cut down the test time for test engineers.

To keep up with demand, ATE manufacturers have graduated to an advanced version of the existing PXI platform. Named PXI Express (PXIe), it increases testing speed and bandwidth while eliminating unnecessary costs.

"Use of PXI Express has increased the speed and throughput of our test equipment by 50 per cent on an average," says Sadaf Arif Siddiqui, business development manager at

Keysight Technologies, in support of PXIe.

Let us see how PXIe platform is charting the success for ATE in diverse applications.

RF test equipment

RF device manufacturers need equipment that can quicken the process of testing their devices for performance, accuracy and traceability of the components. Major players in ATE market such as National Instruments (NI), Rohde and Schwarz, Keysight Technologies and Chroma ATE Inc. have responded to this demand by upgrading their RF test instruments to PXI technology. They are now working towards increasing the bandwidth while reducing the footprint.

NI has developed its second-generation vector signal transceiver (VST) in PXI platform. Talking about it, Avichal Kulshreshtha, technical marketing staff member at NI, says, "Second-generation VST is a compact hardware with field-programmable gate array (FPGA) IC. This IC is embedded with transceiver functions, which are run by the company's LabVIEW software. Test measurement algorithms, bus speeds and CPU speeds have reduced test times, but as RF functions get more complex, improvements are being done."

NI claims that its recently released second-generation VST has five times the bandwidth, 33 per cent smaller footprint and larger user-programmable FPGA than its predecessor. Its instantaneous bandwidth of 1 GHz enables test systems for advanced applications like automotive radars, digital pre-distortion, 5G and 802.11ax.

Similarly, Keysight has developed Keysight Measurement Framework (KMF) for its RF test equipment. The new platform increases the test efficiency while lowering the test time.

"KMF is an easy-to-use software framework for rapid test solution development

